Neil Barot

(908) 448-4557 | neil.barot.jobspace@gmail.com | US Citizen | linkedin.com/in/neilbarot5/ | github.com/Neil1355/

EDUCATION:

Rutgers University-New Brunswick **September 2023 – May 2027**

- Bachelor's in computer science (B.S.) & Minor in Cognitive Science
- Relevant Coursework: DSA, Linear Algebra, Calculus-I & II, Discrete Mathematics. Computer Architecture, Data Management, Cognition and Decision Making, Systems Programming.

PROJECTS:

ProMetric – Tutoring Management System | Python, MySQL, Tkinter **June 2025 – Sept 2025**

- **Developed** a **Python/Tkinter and MySQL desktop application** to manage **30+** students and tutors, automating scheduling and feedback collection and reducing manual tracking time by **~40%**.
- **Designed** a **multi-role dashboard system** for managers, tutors, and students with modules for registration, feedback, and improvement tracking, improving data accessibility and reporting efficiency by **~50%**.

Dunkin Demand Intelligence – Forecasting Platform | Python, Flask, PostgreSQL, React/TS **Nov 2025 - Feb 2026**

- **Engineered** a **full-stack forecasting system** (React, Flask, PostgreSQL) processing **1,000+** daily data points to automate inventory planning and reduce overproduction waste.
- **Implemented** a Seasonal Naive Weekday Model using a **short rolling window mean** (n ≤4) to establish a baseline demand for **40+** specific product SKUs.
- **Developed** a **Multiplicative Bias Layer** that applies **heuristic weights** (0.8, 1.0, 1.2) based on manager-inputted environmental context to refine **prediction accuracy.**
- **Optimized database architecture** using **connection pooling,** resulting in a **100–300ms** reduction in latency and architected a secure **JWT-based** multi-role access control system.

RELEVANT EXPERIENCE:

Barista | Dunkin, New Brunswick, NJ **Nov 2025 - Present**

- **Optimized station setups** and coordinated with a **team of 4-6** to improve order throughput by **20%** while serving **150-250** customers per shift.
- Managed daily inventory for **40+ SKUs** , identifying low-stock patterns and reducing product shortages by **15%** through proactive restocking.

Product Data Research Intern | H15DEN, Long Beach,CA **Nov 2025 – Jan 2026**

- **Performed Exploratory Data Analysis** (EDA) on **300+ data points** across multiple cities to improve **dataset reliability** and enable smoother integration into **early product prototypes.**
- **Collaborated** with a **team of 52** to , reducing inconsistencies by **40%** and improving lead-processing logic efficiency by **30%.**

Designated Hitter – Store Associate + Pharmacy Technician | Walgreens, NJ, **Feb 2025 – Sept 2025**

- **Facilitated 300–400** customer interactions per shift across **retail and pharmacy** workflows, diagnosing issues and routing complex cases to ensure consistent service quality.
- **Resolved data-entry, inventory, and POS discrepancies** with **99% accuracy**, collaborating with pharmacists to **standardize task flows** and reduce operational bottlenecks.

SKILLS & LANGUAGES:

- **Languages**: Python, Java, TypeScript, SQL, HTML/CSS
- **Machine Learning & Data:** scikit-learn, NLTK, spaCy, Hugging Face Transformers, Time-Series Analysis, Heuristic Weighting.
- **Frameworks/Librairies**: Flask, React, NumPy, Pandas, Tkinter.
- **Database & Tools:** PostgreSQL, MySQL, Git/GitHub, Google Cloud, Vercel, REST APIs.
- **Backend & Security**: JWT Authentication, Bcrypt, Connection Pooling, Database Optimization, Audit Logging.